

Head Circumference Estimation from Ultrasound Images Using Convolutional Neural Networks

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Abstract

Ultrasound imaging is widely used to monitor fetal development during pregnancy. One important measurement is the head circumference (HC), which helps doctors evaluate the growth of the fetus. In this study, a convolutional neural network (CNN) is used to estimate the head circumference directly from ultrasound images. The model is trained and evaluated using Mean Absolute Error (MAE). The experimental results show that the proposed method can predict head circumference with good accuracy.

I. Introduction

Medical imaging plays an important role in prenatal healthcare. Ultrasound images are commonly used because they are safe and non-invasive. Head circumference is a key indicator of fetal growth and brain development. Traditionally, this measurement is performed manually by medical experts.

Recently, deep learning methods have shown strong performance in medical image analysis. Therefore, this study aims to use a convolutional neural network to automatically predict the head circumference from ultrasound images. This approach can help reduce manual effort and improve measurement consistency.

II. Dataset Description

The dataset used in this study consists of ultrasound images and corresponding head circumference values. The data is divided into a training set and a validation set.

A. Data Structure

Each data sample includes:

- An ultrasound image file.
- Pixel size information in millimeters.
- The ground truth head circumference value in millimeters.

A total of 999 images are provided, where 799 images are used for training and 200 images are used for validation.

B. Summary Statistics

The head circumference values range from approximately 44 mm to 346 mm. Most samples are concentrated around the middle range, which represents normal fetal development stages.

C. Data Distribution

Figure 1 shows the distribution of head circumference values in the dataset. The histogram indicates that the data is not uniformly distributed. However, it still covers a wide range of values, which is useful for training a regression model.

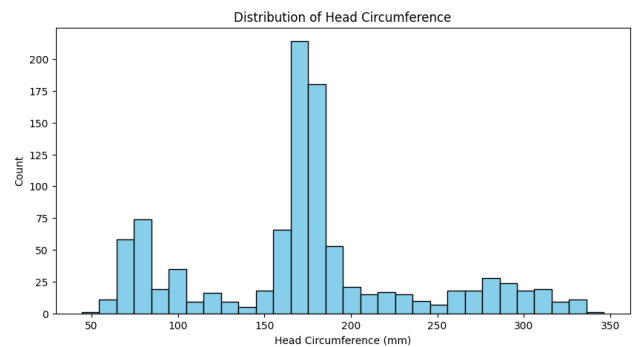


Figure 1: Distribution of head circumference values

III. Implementation

The model is implemented using Python and the PyTorch deep learning framework. Google Colab is used for training with GPU acceleration.

A. Preprocessing

All images are resized and normalized before being fed into the network. The dataset is loaded using a custom PyTorch Dataset class. The training set contains 799 images, while the validation set contains 200 images.

B. Model Selection

A convolutional neural network based on a pretrained ResNet architecture is used. The pretrained weights help the model learn faster and improve performance. The final fully connected layer is modified to output a single continuous value for head circumference prediction.

IV. Training Setup

The model is trained using the Mean Absolute Error (MAE) loss function. The Adam optimizer is applied to update the model parameters. Training is performed for multiple epochs until the validation error stabilizes.

The dataset split is shown below:

- Training set: 799 images
- Validation set: 200 images

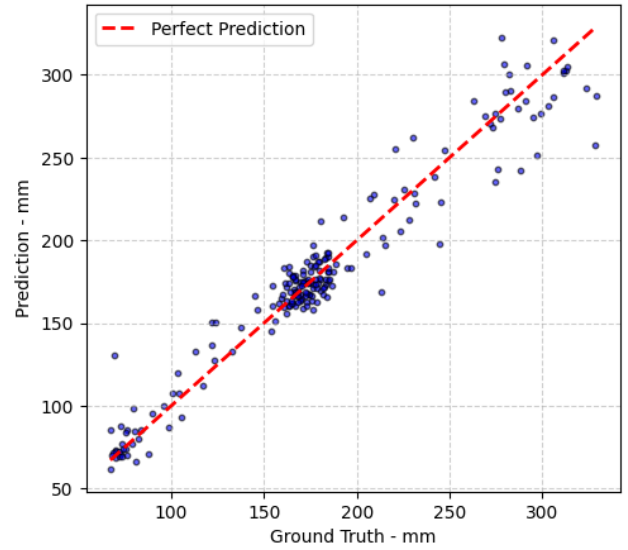


Figure 2: Predicted head circumference versus ground truth

Although the results are good, some prediction errors still exist. This may be caused by image noise, variation in fetal position, or limited data.

V. Results and Discussion

After training, the model is evaluated on the validation set. The final Mean Absolute Error obtained is 10.97 mm.

Figure 2 shows the relationship between predicted values and ground truth values. Most points are close to the diagonal line, which represents perfect prediction. This indicates that the model is able to learn the relationship between ultrasound images and head circumference.

VI. Conclusion

This study demonstrates that a convolutional neural network can be used to estimate fetal head circumference from ultrasound images. The proposed model achieves a validation MAE of 10.97 mm, which shows good prediction accuracy.

In the future, performance can be improved by using more training data, stronger data augmentation, or more advanced neural network architectures. This approach has the potential to support doctors in prenatal healthcare.